

A **subrepresentation** of a representation V of the magnetic point group (G, ϕ) is a subspace $W \subset V$ such that the restriction of the G -action to W is also a representation of (G, ϕ) . A representation V of (G, ϕ) is **irreducible** if its only representations are V and 0 .

Now, the category of representations of finite magnetic groups is semisimple.

Proposition 1.9. *Every representation of a finite magnetic group is a sum of irreducible representations.*

Which follows from Maschke's lemma applied to representations of magnetic groups.

Lemma 1.10 (Maschke). *If W is a subrepresentation of V of a finite magnetic group, then there exists a subrepresentation W' of V such that $V = W \oplus W'$.*

Proof. Write V as a direct sum $V = W \oplus U$ (with U not necessary a subrepresentation) and consider the projection $\pi_1 : V \rightarrow W$ given by $\pi_1(w + u) = w$. Define

$$(1.10) \quad \begin{aligned} \pi : V &\rightarrow W \\ v &\mapsto \sum_{g \in G} g \cdot \pi_1(g^{-1} \cdot v). \end{aligned}$$

The morphism π is in $\text{Hom}_{\mathbf{Rep}(G, \phi)}(V, W)$ because if $h \in G$ and $v \in V$, then

$$(1.11) \quad \pi(h \cdot v) = \sum_{g \in G} g \cdot \pi_1(g^{-1} \cdot (h \cdot v))$$

$$(1.12) \quad = \sum_{g \in G} g \cdot \pi_1((g^{-1}h) \cdot v)$$

$$(1.13) \quad = \sum_{g \in G} (hg^{-1}) \cdot \pi_1((g^{-1}h)^{-1} \cdot v)$$

$$(1.14) \quad = h \cdot \sum_{g \in G} (g^{-1}h) \cdot \pi_1((g^{-1}h)^{-1} \cdot v)$$

$$(1.15) \quad = h \cdot \sum_{l \in G} l \cdot \pi_1(l^{-1} \cdot v)$$

$$(1.16) \quad = h \cdot \pi(v)$$

The morphism π is surjective, this follows since $\pi_1(w) = w$ for $w \in W$, so $\pi(w) = |G|w$. If we denote by $W' = \ker \pi$, then we have the decomposition as a direct sum of representations of (G, ϕ)

$$(1.17) \quad V \cong \text{Im } \pi \oplus \ker \pi = W \oplus W'.$$

□

One key step required to classify the irreducible representations of magnetic groups (G, ϕ) is to construct the induced representations of the core $G_0 \leq G$.

Definition 1.11. Let (G, ϕ) be a finite magnetic point group and V a representation of the core group G_0 . The **induced representation** of V is the complex vector space

$$(1.18) \quad \text{Ind}_{G_0}^{(G, \phi)} V := \mathbb{C}[G, \phi] \otimes_{\mathbb{C}[G_0]} V$$